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Seokju Cho Ph.D. Student

Intro.

I am a PhD student at KAIST and a research intern at NVIDIA. My research focuses on general video perception, with an emphasis on Track Any Point (point tracking) and its applications to 3D and 4D video understanding and reconstruction. I have had the opportunity to intern at NVIDIA and Adobe, where I was honored to work with inspiring mentors, including Orazio Gallo, Abhishek Badki, Hang Su, and Jindong Jiang at NVIDIA, as well as Joon-Young Lee and Gabriel Huang at Adobe.

Internships

NVIDIA Research | Taiwan (Remote) Jan. 2026 - Now
- Mentors: Ryo Hachiuma and Min-Hung Chen

NVIDIA Research | Santa Clara, CA, USA Jun. 2025 - Nov. 2025
- Mentors: Orazio Gallo, Abhishek Badki, Hang Su, and Jindong Jiang.

Adobe Research | San Jose, CA, USA Jun. 2024 - Sep. 2024
- Mentors: Gabriel Huang and Joon-Young Lee.

Adobe Research | San Jose, CA, USA Jun. 2023 - Sep. 2023
- Mentors: Joon-Young Lee and Gabriel Huang.

Publications

1. NVIDIA (incl. **Seokju Cho**), "Nemotron 3.5 Super and Lightning: Open, Efficient Mixture-of-Experts Hybrid Mamba-Transformer Models for Agentic AI",
NVIDIA Technical Report, 2026. [Paper] [Blog]
2. **Seokju Cho**, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, and Min-Hung Chen, "SpatialClaw: Rethinking Action Interface for Agentic Spatial Reasoning",
Neural Information Processing Systems (**NeurIPS**), 2026. [Link]
3. **Seokju Cho**, Abhishek Badki, Hang Su, Jindong Jiang, Ziyao Zeng, Seungryong Kim, Sifei Liu, and Orazio Gallo, "4DP-QA: Scalable QA for 4D Perception in Vision Language Models",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), 2026.
4. Jahyeok Koo, Inès Hyeonsu Kim, Mungyeom Kim, Junghyun Park, Seohyeon Park, Jaeyeong Kim, Jung Yi, **Seokju Cho**, and Seungryong Kim, "MV-TAP: Tracking Any Point in Multi-View Videos",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), 2026.
5. Inès Hyeonsu Kim*, **Seokju Cho***, Jahyeok Koo, Junghyun Park, Jiahui Huang, Honglak Lee, Joon-Young Lee, and Seungryong Kim, "AnthroTAP: Learning Point Tracking with Real-World Motion",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), 2026.
6. Inès Hyeonsu Kim, Woojeong Jin, Soowon Son, Junyoung Seo, **Seokju Cho**, JeongYeol Baek, Byeongwon Lee, JoungBin Lee, and Seungryong Kim, "Pose-dIVE: Pose-Diversified Augmentation for Person Re-Identification",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**) Findings, 2026.
7. **Seokju Cho**, Jiahui Huang, Seungryong Kim, and Joon-Young Lee, "Seurat: From Moving Points to Depth",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), **Highlight**, 2025. [Link]
8. Inès Hyeonsu Kim*, **Seokju Cho***, Jiahui Huang, Jung Yi, Joon-Young Lee, and Seungryong Kim, "Exploring Temporally-Aware Features for Point Tracking",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), 2025. [Link]
9. Kihong Kim*, Yunho Kim*, **Seokju Cho**, Junyoung Seo, Jisu Nam, Kychul Lee, Seungryong Kim, and KwangHee Lee, "DiffFace: Diffusion-based Face Swapping with Facial Guidance",
Pattern Recognition (**PR**), 2025. [Link]
10. Sangbeom Lim*, Seongchan Kim*, Seungjun An*, **Seokju Cho**, Paul Hongsuck Seo, Seungryong Kim, "Multi-Granularity Video Object Segmentation",
AAAI Conference on Artificial Intelligence (**AAAI**), 2025. [Link]
11. Heeseong Shin, Chaehyun Kim, Sunghwan Hong, **Seokju Cho**, Anurag Arnab, Paul Hongsuck Seo, Seungryong Kim, "Towards Open-Vocabulary Semantic Segmentation Without Semantic Labels",
Neural Information Processing Systems (**NeurIPS**), 2024. [Link]

13. **Seokju Cho**, Jiahui Huang, Jisu Nam, Honggyu An, Seungryong Kim, and Joon-Young Lee, "Local All-Pair Correspondence for Point Tracking",
European Conference on Computer Vision (**ECCV**), 2024. [Link]

14. **Seokju Cho**, Jiahui Huang, Seungryong Kim, and Joon-Young Lee, "FlowTrack: Revisiting Optical Flow for Long-Range Dense Tracking",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), 2024. [Link]

15. **Seokju Cho***, Heeseong Shin*, Sunghwan Hong, Anurag Arnab, Paul Hongsuck Seo, and Seungryong Kim, "CAT-Seg: Cost Aggregation for Open-Vocabulary Semantic Segmentation",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), **Highlight**, 2024. [Link]

16. Sunghwan Hong*, **Seokju Cho***, Seungryong Kim, Stephen Lin, "Unifying Feature and Cost Aggregation with Transformers for Dense Correspondence",
International Conference on Learning Representations (**ICLR**), 2024. [Link]

17. Jiuhn Song*, Seonghoon Park*, Honggyu An*, **Seokju Cho**, Min-Seop Kwak, Sungjin Cho, and Seungryong Kim, "DäRF: Boosting Radiance Fields from Sparse Inputs with Monocular Depth Adaptation",
Neural Information Processing Systems (**NeurIPS**), 2023. [Link]

18. Jihye Park*, Sunwoo Kim*, Soohyun Kim*, **Seokju Cho**, Jaejun Yoo, Youngjung Uh, and Seungryong Kim, "LANIT: Language-Driven Image-to-Image Translation for Unlabeled Data",
IEEE Conference on Computer Vision Pattern Recognition (**CVPR**), 2023. [Link]

19. Junyoung Seo*, Gyuseong Lee*, **Seokju Cho**, Jiyoung Lee, Seungryong Kim, "MIDMs: Matching Interleaved Diffusion Models for Exemplar-based Image Translation",
AAAI Conference on Artificial Intelligence (**AAAI**), 2023. [Link]

20. **Seokju Cho***, Sunghwan Hong*, Seungryong Kim, "CATs++: Boosting Cost Aggregation with Convolutions and Transformers",
IEEE Trans. on Pattern Analysis and Machine Intelligence (**TPAMI**), 2023. [Link]

21. Sunghwan Hong, Jisu Nam, **Seokju Cho**, Susung Hong, Sangryul Jeon, Dongbo Min, and Seungryong Kim, "Neural Matching Fields: Implicit Representation of Matching Fields for Visual Correspondence",
Neural Information Processing Systems (**NeurIPS**), 2022. [Link]

22. Sunghwan Hong*, **Seokju Cho***, Jisu Nam, Stephen Lin, Seungryong Kim, "Cost Aggregation with 4D Convolutional Swin Transformer for Few-Shot Segmentation",
European Conference on Computer Vision (**ECCV**), 2022. [Link]

23. **Seokju Cho***, Sunghwan Hong*, Sangryul Jeon, Yunsung Lee, Kwanghoon Sohn, Seungryong Kim, "CATs: Cost Aggregation Transformers for Visual Correspondence",
Neural Information Processing Systems (**NeurIPS**), 2021. [Link]

Awards and Honors

- **Qualcomm Innovation Fellowship Finalist,** 2025.11
- **Google East Asia Student Travel Grants for CVPR,** 2024.06
- **3rd Place Award in AI Online Competition**, Ministry of Science and ICT & National IT Industry Promotion Agency, *Won 300M KRW*, 2023.05

Academic Services

Area Chair: NeurIPS 2026 Evaluations and Datasets Track (**Top Area Chair**).
Reviewers for: CVPR (2023–2026), ECCV (2024), ICCV (2025), NeurIPS (2023).

Education

Korea Advanced Institute of Science and Technology (KAIST) Seoul, Korea
Integrated M.S./Ph.D. in Artificial Intelligence 2024 - 2027 *(expected)*

Korea University Seoul, Korea
Integrated M.S./Ph.D. in Computer Science and Engineering 2022 - 2024
- Transferred to KAIST with supervisor (degree incomplete).

Yonsei University Seoul, Korea
B.S. in Computer Science 2018 - 2022